

Lattice Theory (1)

CSE552 Program Analysis — Lecture 5

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Sign Analysis

- ▶ An analysis that finds out the possible integer values of variables and expressions
- ▶ In concrete executions, values can be arbitrary integers
- ▶ Our analysis considers an abstraction of the integer values by grouping them into the three categories, or abstract values: negative ($-$), zero (0), and positive ($+$)
- ▶ e.g., if a sound analysis concludes that the final value of a variable is $+$, then the value must be a positive integer in any execution

- ▶ The analysis may not know the sign of some expression
 - ▶ The value is positive in some execution, and it is not in other executions
 - ▶ The value is positive in every execution but the analysis fails to recognize this (which is unavoidable due to undecidability)
- ▶ We add a special abstract value $\top$ representing “don’t know”

Bottom

- ▶ It is beneficial to also have an abstract value $\perp$ for expressions
 - ▶ whose values are not numbers (e.g., pointers), or
 - ▶ that have no value in any execution because they are unreachable

Sign Analysis Example

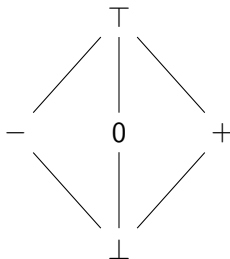
```
a = 42;  
b = 87;  
if input() {  
    c = a + b;  
} else {  
    c = a - b;  
}
```

A sound analysis may conclude that, at return,

- ▶ a is $+$
- ▶ b is $+$
- ▶ c is $\top$

Abstract Domains

- ▶ For this analysis, we have an abstract domain consisting of the five abstract values: $\{\perp, -, 0, +, \top\}$
- ▶ We can organize as follows with the least precise information at the top and the most precise information at the bottom:



Questions — Motivation for Lattice Theory

- ▶ How can we ensure termination and soundness of our analysis?
- ▶ What is required for the abstract domain?
- ▶ We need a mathematical foundation: the lattice theory
- ▶ The connection between lattices and program analysis was established in the seminal work by Kildall, Kam, and Ullman¹²

¹A unified approach to global program optimization (Kildall, 1973)

²Monotone data flow analysis frameworks (Kam and Ullman, 1977)

Partially Ordered Sets — Definition

Definition (Partially ordered set). A partially ordered set (poset) is a set S equipped with a binary relation $\sqsubseteq$ where the following conditions are satisfied:

- ▶ Reflexivity: $\forall x \in S. x \sqsubseteq x$
- ▶ Transitivity: $\forall x, y, z \in S. x \sqsubseteq y \wedge y \sqsubseteq z \Rightarrow x \sqsubseteq z$
- ▶ Anti-symmetry: $\forall x, y \in S. x \sqsubseteq y \wedge y \sqsubseteq x \Rightarrow x = y$

c.f., a total order additionally requires totality
($\forall x, y \in S. x \sqsubseteq y \vee y \sqsubseteq x$)

Partially Ordered Sets — Intuition and Notation

- ▶ From the analysis perspective, when $x \sqsubseteq y$, we say that “ y is a safe approximation of x ,” or that “ x is at least as precise as y ”
- ▶ We sometimes write $y \sqsupseteq x$ instead of $x \sqsubseteq y$

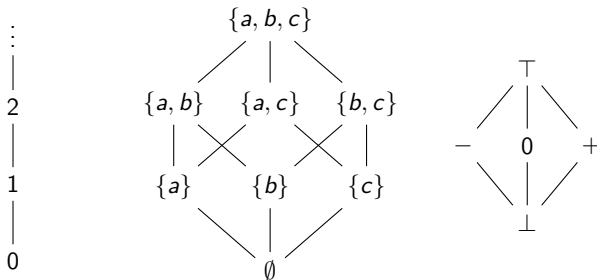
Partially Ordered Sets — Examples

Examples:

- ▶ $(\mathbb{N}, \leq)$
- ▶ $(\mathcal{P}(S), \subseteq)$
- ▶ $(\{\perp, -, 0, +, \top\}, \sqsubseteq)$ where
 - ▶ $\perp \sqsubseteq \perp, -, 0, +, \top$
 - ▶ $- \sqsubseteq -, \top$
 - ▶ $0 \sqsubseteq 0, \top$
 - ▶ $+ \sqsubseteq +, \top$
 - ▶ $\top \sqsubseteq \top$

Hasse Diagrams

- ▶ A partial order can be illustrated by a Hasse diagram in which the elements are nodes and the order relation is the transitive closure of edges leading from lower to higher nodes



Bounds

Definition (Upper bound and lower bound). For $X \subseteq S$ and $y \in S$,

- ▶ y is an upper bound for X , written $X \sqsubseteq y$, if $\forall x \in X. x \sqsubseteq y$
- ▶ y is a lower bound for X , written $y \sqsubseteq X$, if $\forall x \in X. y \sqsubseteq x$

Example: Given $(\mathbb{N}, \leq)$, for $\{5, 7, 10\}$, 10 and 100 are some of upper bounds, and 1 and 5 are some of lower bounds.

Least Upper Bound and Greatest Lower Bound

Definition (Least upper bound and greatest lower bound).

- ▶ A least upper bound (lub), written $\sqcup X$, satisfies

$$X \sqsubseteq \sqcup X \wedge \forall y. X \sqsubseteq y \Rightarrow \sqcup X \sqsubseteq y$$

- ▶ A greatest lower bound (glb), written $\sqcap X$, satisfies

$$\sqcap X \sqsubseteq X \wedge \forall y. y \sqsubseteq X \Rightarrow y \sqsubseteq \sqcap X$$

Lub and Glb — Notations

Notations:

- ▶ $x \sqcup y := \sqcup\{x, y\}$ (join of x and y)
- ▶ $x \sqcap y := \sqcap\{x, y\}$ (meet of x and y)
- ▶ $\sqcup_{a \in A} f(a) := \sqcup\{f(a) \mid a \in A\}$
- ▶ $\sqcap_{a \in A} f(a) := \sqcap\{f(a) \mid a \in A\}$

Lub and Glb — Properties

Important properties:

- ▶ If $\sqcup X$ exists, then it is unique
- ▶ If $\sqcap X$ exists, then it is unique
- ▶ If $x \sqcup y$ exists, then $x \sqsubseteq y \iff x \sqcup y = y$
- ▶ If $x \sqcap y$ exists, then $x \sqsubseteq y \iff x \sqcap y = x$

Lub and Glb — Examples and Role in Analysis

Examples:

- ▶ In $(\mathbb{N}, \leq)$, $\sqcup = \max$ and $\sqcap = \min$
- ▶ In $(\mathcal{P}, \subseteq)$, $\sqcup = \cup$ and $\sqcap = \cap$
- ▶ The lub/join operation plays an important role in program analysis
- ▶ We use lub when combining abstract information from multiple sources
 - ▶ e.g., when control flow merges after the branches of if statements

Lattices — Definition

Definition (Lattice).

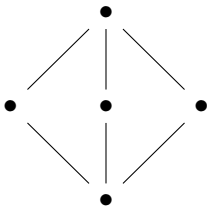
- ▶ A lattice is a partial order $(S, \sqsubseteq)$ in which $x \sqcup y$ and $x \sqcap y$ exist for all $x, y \in S$
- ▶ A complete lattice is a partial order $(S, \sqsubseteq)$ in which $\bigsqcup X$ and $\bigsqcap X$ exist for all $X \subseteq S$

Lattices — Properties

Important properties:

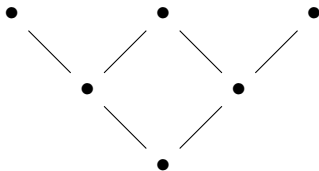
- ▶ Every complete lattice is a lattice
 - ▶ What is a lattice that is not a complete lattice?
- ▶ A nonempty finite lattice is complete
- ▶ Where S is a poset, every subset of S has an lub $\iff$ every subset of S has a glb
- ▶ Most lattices we encounter in program analysis are complete lattices

Lattice Example 1



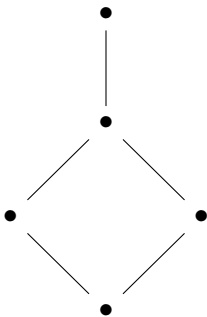
Lattice

Lattice Example 2



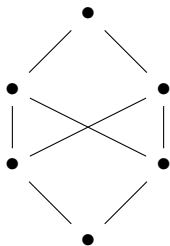
Not a lattice

Lattice Example 3



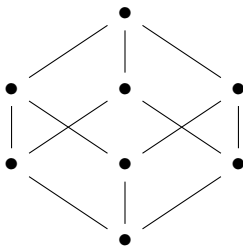
Lattice

Lattice Example 4



Not a lattice

Lattice Example 5



Lattice

Top and Bottom

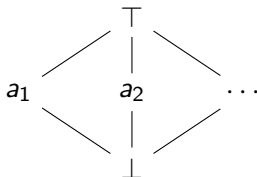
- ▶ $\top = \bigsqcup S$
- ▶ $\perp = \bigsqcap S$
- ▶ Height of S : the length of the longest path from $\perp$ to $\top$
 - ▶ e.g., the height of the sign lattice is 2
- ▶ Every complete lattice has $\top$ and $\perp$

Constructing Lattices — Power Sets

- ▶ $(\mathcal{P}(A), \subseteq)$ is a complete lattice, called the power set lattice:
 - ▶ $\sqcup = \cup, \quad \sqcap = \cap, \quad \perp = \emptyset, \quad \top = A$
- ▶ $(\mathcal{P}(A), \supseteq)$ is also a complete lattice, called the reverse power set lattice:
 - ▶ $\sqcup = \cap, \quad \sqcap = \cup, \quad \perp = A, \quad \top = \emptyset$

Constructing Lattices — Flat

- ▶ For $A = \{a_1, a_2, \dots\}$, $\text{flat}(A)$ is a complete lattice with height 2:



Example:

- ▶ $\text{Sign} = \{\perp, -, 0, +, \top\}$ can be expressed as $\text{flat}(\{-, 0, +\})$

Constructing Lattices — Products (Definition)

- ▶ If $L_1, L_2, \dots, L_n$ are complete lattices, then so is the product where the order $\sqsubseteq$ is defined componentwise:
 - ▶ $L_1 \times L_2 \times \dots \times L_n = \{(x_1, x_2, \dots, x_n) \mid x_i \in L_i\}$
 - ▶ $(x_1, \dots, x_n) \sqsubseteq (x'_1, \dots, x'_n) \iff \forall i. x_i \sqsubseteq x'_i$
- ▶ Product of n identical lattices can be written concisely as $L^n = L \times L \times \dots \times L$
- ▶ Lubs and glbs can be computed componentwise:

$$\bigsqcup_{i \in I} (x_{i1}, \dots, x_{in}) = \left(\bigsqcup_{i \in I} x_{i1}, \dots, \bigsqcup_{i \in I} x_{in} \right)$$
$$\bigsqcap_{i \in I} (x_{i1}, \dots, x_{in}) = \left(\bigsqcap_{i \in I} x_{i1}, \dots, \bigsqcap_{i \in I} x_{in} \right)$$

Constructing Lattices — Products (Height and Examples)

► $\text{height}(L_1 \times \cdots \times L_n) = \text{height}(L_1) + \cdots + \text{height}(L_n)$

Examples:

► $\text{Sign} \times \text{Flat}(\mathbb{N})$

► $(+, 1)$

► $(-, \top)$

► $\text{Sign}^{\#(\text{Var})}$

► $(+, 0, \top)$

► $(\top, \top, -)$

Constructing Lattices — Maps (Definition)

- ▶ If A is a set and L is a complete lattice, then we obtain a complete lattice called a map lattice $A \rightarrow L$ consisting of the set of functions from A to L , ordered pointwise:
 - ▶ $A \rightarrow L = \{[a_1 \mapsto x_1, a_2 \mapsto x_2, \dots] \mid \{a_1, a_2, \dots\} = A \wedge x_1, x_2, \dots \in L\}$
 - ▶ $f \sqsubseteq g \iff \forall a \in A. f(a) \sqsubseteq g(a)$
- ▶ Lubs and glbs can be computed pointwise:
 - ▶ $\bigsqcup_{i \in I} f_i = \lambda a. \bigsqcup_{i \in I} f_i(a)$
 - ▶ $\bigsqcap_{i \in I} f_i = \lambda a. \bigsqcap_{i \in I} f_i(a)$
- ▶ $\text{height}(A \rightarrow L) = |A| \cdot \text{height}(L)$

Constructing Lattices — Maps (Examples)

Examples:

- ▶ $\text{Var} \rightarrow \text{Sign}$
- ▶ $\text{CFG Node} \rightarrow \text{Var} \rightarrow \text{Sign}$

Homomorphism and Isomorphism — Definitions

Definition (Homomorphism). Where L_1 and L_2 are lattices, a function $f : L_1 \rightarrow L_2$ is a homomorphism if

$$\forall x, y \in L_1. f(x \sqcup y) = f(x) \sqcup f(y) \wedge f(x \sqcap y) = f(x) \sqcap f(y)$$

Definition (Isomorphism).

- ▶ If a homomorphism f is bijective, f is an isomorphism
- ▶ If there exists an isomorphism $f : L_1 \rightarrow L_2$, L_1 and L_2 are isomorphic, written $L_1 \cong L_2$
- ▶ Intuitively, isomorphic lattices are exactly the same (e.g., same Hasse diagram), only with different names for elements

Homomorphism and Isomorphism — Examples

Examples:

- ▶ $L^n \cong A \rightarrow L$ where $\#(A) = n$
- ▶ $\text{Sign}^{\#(\text{Var})} \cong \text{Var} \rightarrow \text{Sign}$

Summary

- ▶ Sign analysis abstracts integer values into $\{-, 0, +\}$ with $\top$ (unknown) and $\perp$ (no value)
- ▶ A complete lattice is a poset where every subset has an lub and a glb
- ▶ The lub (join) operation combines abstract information at control flow merge points
- ▶ Complete lattices can be constructed using power sets, flat, products, and maps